Gaussian Rigidity from Symmetrized Moment Generating Functions: Chi-Square Quadratic Forms and Sharp Reflection Stability

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Abstract

We study Gaussian rigidity through the symmetrized moment generating function $M_X(s)M_X(-s)$. For independent centered coordinates $X = (X_i)$ and a positive-semidefinite matrix A, we prove that if $X^\mathsf{T} A X$ has a chi-square distribution and every active coordinate satisfies the local dominance $M_i(s)M_i(-s) \geq \exp(\sigma_i^2 s^2)$, then all active coordinates are Gaussian. We then turn to the quantitative factor-recovery problem. Under a fixed exponential-moment bound, if $\log M_X(s) + \log M_X(-s) - s^2$ is uniformly bounded in absolute value by Δ on a fixed neighborhood of the origin, then the Kolmogorov distance from X to the standard normal law is at most

$$C\sqrt{\frac{\log\log(1/\Delta)}{\log(1/\Delta)}}.$$

The order is optimal in general: a Laguerre perturbation yields a matching lower bound. The proof converts local reflected-MGF information into high-order moments, produces a growing zero-free disk for a truncated characteristic function, and then applies logarithmic coefficient bounds and Esseen smoothing. As applications, we obtain exact and quantitative results for the sample variance; a lattice example shows that no analogous total-variation stability can hold.

Keywords: Gaussian characterization; symmetrized moment generating function; chi-square quadratic forms; sample variance; Cramér stability; zero-free characteristic functions

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1 Introduction and main results

The Gaussian decomposition principle used in the factor-recovery argument is the classical theorem of Cramér [1].

The exact chi-square law of the sample variance is one of the classical signatures of Gaussian samples. The unrestricted converse for a fixed sample size $n \geq 3$ remains a longstanding characterization problem. Existing results impose additional structure, including symmetry or infinite divisibility; see [2–7]. We do not claim to resolve that unrestricted problem. The sample variance is a chi-square quadratic form, and spherical averaging of its Laplace transform naturally brings in the symmetrized quantity $M_X(s)M_X(-s)$.

For a centered variance-one random variable X, let $M_X(s) = \mathbb{E}e^{sX}$ whenever finite and define

$$R_X(s) := \log M_X(s) + \log M_X(-s) - s^2, \qquad \Delta_\rho(X) := \sup_{|s|\leq\rho} |R_X(s)|. \tag{1}$$

We write $d_{\mathrm{K}}(X, Z)$ for the Kolmogorov distance between the laws of X and Z,

$$d_{\mathrm{K}}(X, Z) := \sup_x |\mathbb{P}(X \leq x) - \mathbb{P}(Z \leq x)| .$$

We work under the fixed exponential envelope

$$\mathbb{E}e^{2\tau|X|} \leq K. \tag{2}$$

If X' is an independent copy of X, then $M_{X-X'}(s) = M_X(s)M_X(-s)$. The difficulty is that reflection removes the Fourier phase. Our stability theorem shows that the factor can nevertheless be recovered, at the optimal logarithmic scale, under (2).

The proof uses two scales. A fixed complex neighborhood yields high-order moment information for $X - X'$; after truncation, this produces a Gaussian product approximation on a disk of radius of order $\sqrt{m}$. Zero-freeness on this disk allows one to take a logarithm and recover the factor. The exact theorem, by contrast, follows from equality in Jensen's inequality after spherical averaging.

Exact rigidity

The positive-semidefinite quadratic-form theorem below motivates the reflection condition.

Theorem 1.1 (Exact PSD rigidity) *Let $X = (X_1, \ldots, X_d)$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\mathrm{Var}(X_i) = \sigma_i^2$. Let $A = A^\mathsf{T} \succeq 0$ have rank $r \geq 1$, and suppose*

$$X^\mathsf{T} AX \sim \chi_r^2.$$

For every active coordinate i with $A_{ii} > 0$, assume that $M_i(s) = \mathbb{E}e^{sX_i}$ is finite in a neighborhood of the origin and

$$M_i(s)M_i(-s) \geq e^{\sigma_i^2 s^2}$$

there. Then every active X_i is Gaussian, $X_i \sim \mathcal{N}(0, \sigma_i^2)$.

Characterizations based on quadratic forms have a long history. Converse results for sample variance and related forms use additional structural assumptions; see, for example, Christoph–Prohorov–Ulyanov, Kruglov, and Golikova–Kruglov [5, 6, 8]. The condition here is different: infinite divisibility implies the reflected-MGF dominance through the Lévy–Khintchine formula, but the converse fails and no symmetry is required. For an infinitely divisible centered law with finite variance, the representation gives, for every s such that both $M_X(s)$ and $M_X(-s)$ are finite,

$$\log M_X(s) + \log M_X(-s) - \sigma^2 s^2 = 2 \int \left(\cosh(sx) - 1 - \frac{s^2 x^2}{2} \right) \nu(dx) \geq 0.$$

The converse fails. If X takes the values 3 and $-1/3$ with probabilities $1/10$ and $9/10$, then it is centered and has variance one. It is not infinitely divisible: if a bounded non-degenerate law of support diameter d were an n-fold convolution of an i.i.d. root, that root would have support diameter at most d/n, and Popoviciu's inequality would give $\mathrm{Var}(X)/n \leq d^2/(4n^2)$, which is impossible for large n. Its fourth cumulant is $46/9$, so $R_X(s) = 23s^4/54 + O(s^6)$ and the local dominance holds. This example is asymmetric. In contrast, a Rademacher variable ε satisfies $R_\varepsilon(s) = 2 \log \cosh s - s^2 = -s^4/6 + O(s^6)$, which is negative for all sufficiently small nonzero s.

The Lévy–Khintchine representation used in this comparison is standard; see Sato [9].

Reflection stability

The quantitative result recovers a factor from local reflected-MGF information without assuming the dominance sign.

Theorem 1.2 (Sharp reflection stability) *Fix $\rho, \tau, K > 0$. There exist constants $C < \infty$ and $\Delta_0 \in (0, e^{-e})$, depending only on (ρ, τ, K), such that the following holds. If*

$$\mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1, \qquad \mathbb{E}e^{2\tau|X|} \leq K,$$

and

$$|R_X(s)| \leq \Delta, \qquad |s| \leq \rho,$$

for some $0 < \Delta \leq \Delta_0$, then

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \leq C \sqrt{\frac{\log\log(1/\Delta)}{\log(1/\Delta)}}.$$

Classical Cramér stability starts from a two-factor convolution $F * G$ that is already close to a Gaussian convolution in a global probability metric such as Kolmogorov or Lévy distance. The reflected convolution $F * \widetilde{F}$ is therefore a legitimate special case of that general framework. The equal-factor case $F * F$ retains the Fourier phase and admits stronger iid estimates, as in the theory of Bobkov, Chistyakov, and Götze [10, 11]. In the present problem the available input is different: the quantity $M_X(s)M_X(-s)$ is controlled only through $R_X(s)$ on a fixed real interval, and its Fourier counterpart

$|\varphi_X|^2$ has lost the phase. The main difficulty is not that reflection lies outside Cramér-type theory, but that local transform information must be converted into global factor recovery without phase information.

One could first try to convert the local reflected-MGF control into a global approximation for $X - X'$ and then invoke a general stability theorem for Cramér decomposition. Such an argument discards the structure responsible for the sharp modulus. Our proof instead keeps the reflected factorization throughout: high-order moment information constructs a zero-free disk for a truncated factor itself, after which the logarithm of that factor is controlled directly. The logarithmic coefficient step is a finite-disk estimate in the spirit of quantitative Marcinkiewicz theory; the zero-free disk here is derived from the reflected-MGF analysis rather than assumed, as in the quantitative results of Eremenko and Fryntov and of Dinh, Ghosh, Tran, and Tran [12, 13].

The order in this theorem cannot be improved uniformly in general. The obstruction is supplied by high-order odd perturbations whose first m odd moments vanish.

Theorem 1.3 (Laguerre lower bound) *There are constants $\rho, \tau, K, c > 0$ and centered variance-one random variables W_m such that, for all sufficiently large m,*

$$\mathbb{E}e^{2\tau|W_m|} \le K, \qquad R_{W_m}(s) \ge 0 \quad (|s| \le \rho),$$

and, with

$$\Delta_m := \sup_{|s|\le\rho} R_{W_m}(s) \to 0,$$

$$\log\frac{1}{\Delta_m} = 2m\log m + O(m), \qquad d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) \ge cm^{-1/2}.$$

Consequently, for some $c' > 0$,

$$d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) \ge c'\sqrt{\frac{\log\log(1/\Delta_m)}{\log(1/\Delta_m)}}$$

for all sufficiently large m.

2 Exact rigidity for chi-square quadratic forms

We prove Theorem 1.1 and then apply it to the ordinary sample variance. Write $A = B^\mathsf{T}B$, where $B \in \mathbb{R}^{r\times d}$ has full row rank, and let b_i be the ith column of B. If V is uniform on S^{r-1}, write

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1}.$$

The transform is well defined for all $z \in \mathbb{C}$ because V_1 is bounded. Spherical averaging gives

$$\mathbb{E}_V \prod_{i=1}^{d} M_i\big(t\langle b_i, V\rangle\big) = \mathbb{E}_{X,V} e^{t\langle BX, V\rangle} = \mathbb{E}\Psi_r\big(t\sqrt{X^\mathsf{T}AX}\big). \tag{3}$$

Since $X^\mathsf{T}AX \sim \chi^2_r$, the final quantity in (3) equals

$$\mathbb{E}_{G,V} e^{t\langle G,V\rangle} = e^{t^2/2}, \qquad G \sim \mathcal{N}(0, I_r). \tag{4}$$

All expectations below are legitimate for sufficiently small $|t|$, because the coordinate MGFs exist on a common neighborhood of the origin and the sphere is compact.

More explicitly, each active coordinate may initially have its own neighborhood $|s| < \rho_i$ on which the MGF and the dominance condition are available. Since there are finitely many coordinates, reducing the radius to $\rho_* := \min_i \rho_i > 0$ gives one common interval for the spherical argument.

Define

$$C_i(s) := \log M_i(s) - \frac{\sigma_i^2 s^2}{2}$$

and

$$H_t(v) := \sum_{i=1}^{d} C_i\big(t\langle b_i, v\rangle\big) + \frac{t^2}{2}\left(\sum_{i=1}^{d} \sigma_i^2 \langle b_i, v\rangle^2 - 1\right). \tag{5}$$

The quadratic terms cancel inside the exponential, so (4) implies

$$\mathbb{E}_V e^{H_t(V)} = e^{-t^2/2}\mathbb{E}_V \prod_i M_i(t\langle b_i, V\rangle) = 1.$$

Jensen's inequality therefore gives $\mathbb{E}_V H_t(V) \le 0$.

The mean of the quadratic correction in (5) is zero. Indeed,

$$\mathbb{E}_V \sum_i \sigma_i^2 \langle b_i, V\rangle^2 = \frac{1}{r}\sum_i \sigma_i^2 \|b_i\|^2 = \frac{1}{r}\mathbb{E}(X^\mathsf{T} A X) = 1,$$

where the last equality uses the mean of χ_r^2. Since V and $-V$ have the same law, coordinatewise dominance now gives

$$\begin{aligned}
2\mathbb{E}_V H_t(V) &= \sum_{i=1}^{d} \mathbb{E}_V \left[\log M_i(t\langle b_i, V\rangle) + \log M_i(-t\langle b_i, V\rangle) - \sigma_i^2 t^2 \langle b_i, V\rangle^2\right] \\
&= \sum_{i=1}^{d} \mathbb{E}_V R_i\big(t\langle b_i, V\rangle\big) \ge 0,
\end{aligned}$$

where $R_i(s) = \log M_i(s) + \log M_i(-s) - \sigma_i^2 s^2$. Hence $\mathbb{E}_V H_t(V) = 0$, and every summand in the last display has zero expectation.

For an active coordinate, $A_{ii} = \|b_i\|^2 > 0$. If $r \ge 2$, the random variable $\langle b_i, V\rangle$ has a density that is strictly positive on the interior of $[-\|b_i\|, \|b_i\|]$. Continuity of R_i therefore implies $R_i(s) = 0$ on a neighborhood of the origin. If $r = 1$, then $V \in \{-1, 1\}$ and varying t directly gives the same conclusion, because R_i is even. Thus, for every active coordinate,

$$M_i(s) M_i(-s) = e^{\sigma_i^2 s^2}$$

near the origin. If $\sigma_i^2 = 0$, then $X_i = 0$ almost surely. If $\sigma_i^2 > 0$, an independent copy X_i' gives that $X_i - X_i'$ has the MGF of $\mathcal{N}(0, 2\sigma_i^2)$ near zero and hence has that Gaussian law. The Lévy–Crámer decomposition theorem applied to the independent sum $X_i + (-X_i')$ then implies that X_i is Gaussian.

This proves Theorem 1.1. For Corollary 5.1, take

$$A = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}.$$

It is a rank-$n-1$ orthogonal projection, every diagonal entry is $(n-1)/n > 0$, and its quadratic form is exactly $\sum_i (X_i - \bar{X})^2$. The corollary follows immediately.

3 Reflection stability

Throughout this section let X' be an independent copy of X, set

$$Y = X - X', \qquad G \sim \mathcal{N}(0,2),$$

and define

$$F(z) := M_Y(z) - e^{z^2}, \qquad L := \log(1/\Delta).$$

For probability measures μ and ν, write

$$d_{\mathrm{TV}}(\mu,\nu) := \sup_A |\mu(A) - \nu(A)|.$$

We prove Theorem 1.2 through five lemmas.

3.1 Half-disk propagation

Lemma 3.1 (Two-constants estimate on a half-disk) *Let $D_R^+ = \{z \in \mathbb{C} : |z| < R,\ \Im z > 0\}$. Suppose that F is holomorphic on a neighborhood of $\overline{D_R^+}$ and satisfies*

$$|F(x)| \le a \quad (|x| \le R), \qquad |F(z)| \le b \quad (|z| = R,\ \Im z \ge 0),$$

where $0 < a \le b$, then for every fixed $0 < \eta < 1$ there is $\omega_\eta > 0$, depending only on η, such that

$$\log|F(z)| \le \omega_\eta \log a + (1-\omega_\eta)\log b, \qquad |z| \le \eta R, \quad \Im z \ge 0.$$

The same estimate holds in the lower half-disk.

Proof For $\delta > 0$, the function $u_\delta(z) := \log(|F(z)| + \delta)$ is subharmonic. Its boundary values are at most $\log(a+\delta)$ on the diameter and at most $\log(b+\delta)$ on the semicircle. The two-constants theorem for subharmonic functions therefore bounds u_δ by the corresponding harmonic-measure combination. Under the scaling $z = Rw$, the half-disk D_R^+ becomes the unit half-disk, and the inner region $|z| \le \eta R$ becomes $|w| \le \eta$. The harmonic measure of the diameter is positive on this compact inner region, so its minimum is a constant $\omega_\eta > 0$ depending only on η, not on R, m, or Δ. Letting $\delta \downarrow 0$ gives the assertion. □

3.2 High-order moments of the reflected variable

Lemma 3.2 (Local defect to high-order moments) *Fix ρ, τ, K. For every $A > 0$ there is $c_0 = c_0(A, \rho, \tau, K) > 0$ such that, with*

$$m = \left\lfloor c_0 \frac{L}{\log L} \right\rfloor,$$

all sufficiently small Δ *satisfy*

$$\left|\mathbb{E}Y^k - \mathbb{E}G^k\right| \le e^{-Am\log m}, \qquad 0 \le k \le 2m.$$

In particular,

$$\mathbb{E}|Y|^{2m} \le (Cm)^m.$$

Proof Choose $0 < \rho_0 < \min\{\rho, 2\tau\}$ and put $D^{\pm}_{\rho_0} := \{z : |z| < \rho_0,\ \pm\Im z > 0\}$. Exponential integrability gives a bound $|F(z)| \le B_0$ on the two semicircular arcs, with B_0 depending only on (ρ, τ, K). On the real diameter, the defect hypothesis gives

$$|F(s)| = e^{s^2}|e^{R_X(s)} - 1| \le C_0\Delta, \qquad |s| \le \rho_0,$$

where $F(s) = M_Y(s) - e^{s^2}$ and $Y = X - X'$. For sufficiently small Δ, the diameter bound is no larger than the arc bound. Apply Lemma 3.1 with a fixed $\eta < 1$ to $D^+_{\rho_0}$ and $D^-_{\rho_0}$. For $r_0 := \eta\rho_0$, write $\omega_0 := \omega_\eta$. Thus there are constants $r_0 > 0$, $\omega_0 > 0$, and $C_1 < \infty$ such that

$$\sup_{|z|\le r_0} |F(z)| \le C_1\Delta^{\omega_0}.$$

The Cauchy estimate consequently gives

$$|F^{(k)}(0)| \le C_1 k! r_0^{-k}\Delta^{\omega_0}.$$

For $k \le 2m$, use $\log(k!) \le k\log k$ and $k\log k \le 2m\log(2m)$ to obtain

$$\begin{aligned}\log\bigl(C_1 k! r_0^{-k}\Delta^{\omega_0}\bigr) &\le -\omega_0 L + 2m\log(2m) + 2m|\log r_0| + O(1)\\ &\le -\omega_0 L + O(c_0 L),\end{aligned}$$

when $m = \lfloor c_0 L/\log L\rfloor$ and L is large. Since $m\log m \le c_0 L$ up to a fixed lower-order adjustment, choosing c_0 with $(A+3)c_0 < \omega_0$ absorbs the $O(c_0 L)$ term and gives $|F^{(k)}(0)| \le e^{-Am\log m}$. Since $F^{(k)}(0) = \mathbb{E}Y^k - \mathbb{E}G^k$, the moment conclusion follows. Finally,

$$\mathbb{E}G^{2m} = \frac{(2m)!}{m!} \le (Cm)^m,$$

and the negligible moment mismatch gives the stated $2m$-th moment bound. □

3.3 Tail bound and truncation

Lemma 3.3 (Square-root tail lifting) *There are constants* $C, c > 0$ *with the following property. For every sufficiently large fixed* B*, if* $N = B\sqrt{m}$*, then*

$$\mathbb{P}(|X| > N) \le Ce^{-\beta_B m},$$

where $\beta_B \to \infty$ *as* $B \to \infty$*. If* X_N *denotes the conditional law* $X \mid \{|X| \le N\}$*, then*

$$d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) \le Ce^{-\beta_B m}.$$

Moreover

$$|\mathbb{E}X_N| + |\operatorname{Var}(X_N) - 1| \le C(1+N^2)e^{-cN}.$$

Proof Let a be a median of X. Since $\mathbb{E}X^2 = 1$, $|a| \le \sqrt{2}$. For every $p \ge 1$, conditioning on X and using the median event on the opposite side gives

$$\mathbb{E}|X - a|^p \le 2\mathbb{E}|X - X'|^p.$$

Indeed, if $x \ge a$, then $\mathbb{P}(X' \le a) \ge 1/2$ and $|x - X'| \ge x - a$ on that event; the case $x \le a$ is symmetric. Taking $p = 2m$ and using Lemma 3.2 gives $\|X - a\|_{2m} \le C\sqrt{m}$, and the triangle inequality gives $\|X\|_{2m} \le C\sqrt{m}$. Hence, for $N = B\sqrt{m} > \sqrt{2}$,

$$\mathbb{P}(|X| > N) \le \left(\frac{\|X\|_{2m}}{N}\right)^{2m} \le C\left(\frac{C_0}{B^2}\right)^m.$$

Thus one may take $\beta_B = \frac{1}{2}\log(B^2/C_0)$ for sufficiently large B. If $p_N = \mathbb{P}(|X| > N)$, conditioning gives

$$d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) \le p_N.$$

The exponential envelope also gives

$$\mathbb{E}\big(|X|\mathbf{1}_{\{|X|>N\}}\big) \le Ce^{-cN}, \qquad \mathbb{E}\big(X^2\mathbf{1}_{\{|X|>N\}}\big) \le C(1+N^2)e^{-cN}.$$

Since $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$,

$$|\mathbb{E}X_N| \le \frac{Ce^{-cN}}{1 - p_N}, \qquad |\mathbb{E}X_N^2 - 1| \le \frac{p_N + C(1+N^2)e^{-cN}}{1 - p_N}.$$

It follows that $|\mathbb{E}X_N| + |\operatorname{Var}(X_N) - 1| \le C(1+N^2)e^{-cN}$ after increasing C. □

3.4 Product approximation on a growing interval

Lemma 3.4 (Real-axis product approximation) *For B fixed sufficiently large, there is $a_0 = a_0(B, \rho, \tau, K) > 0$ such that, with*

$$R = a_0\sqrt{m}$$

and φ_N the characteristic function of X_N,

$$\sup_{|t|\le 2R} \left|\varphi_N(t)\varphi_N(-t) - e^{-t^2}\right| \le e^{-\Gamma m}$$

for the finite target exponent $\Gamma > 4$ selected in Appendix A.

Proof Use the $2m - 1$ Taylor polynomial of the characteristic function of $Y = X - X'$ and the real-variable remainder

$$\left|e^{iu} - \sum_{k=0}^{2m-1} \frac{(iu)^k}{k!}\right| \le \frac{|u|^{2m}}{(2m)!}.$$

Taylor's inequality and the moment comparison give

$$|\varphi_Y(t) - e^{-t^2}| \le \frac{|t|^{2m}\mathbb{E}|Y|^{2m}}{(2m)!} + \frac{|t|^{2m}\mathbb{E}|G|^{2m}}{(2m)!} + \sum_{k=0}^{2m-1} \frac{|t|^k}{k!}|\mathbb{E}(iY)^k - \mathbb{E}(iG)^k|,$$

where $G \sim \mathcal{N}(0, 2)$. On $|t| \le 2R$, the two Taylor-remainder terms are bounded by $(Ca_0^2)^m$ after Stirling, while the moment-mismatch term is super-exponentially smaller by Lemma 3.2. Choose a_0 small. Since $\varphi_Y(t) = \varphi_X(t)\varphi_X(-t)$ and the truncation error is bounded by Lemma 3.3, transfer to X_N gives

$$\varphi_N(t)\varphi_N(-t) = \varphi_{X_N - X_N'}(t) = e^{-t^2} + O(e^{-\Gamma m}).$$

The constants in the $O(\cdot)$ term are absorbed by the choice of the fixed target exponent in Appendix A. □

3.5 A zero-free disk

Lemma 3.5 (Growing zero-free disk) *The constants B and a_0 can be chosen so that, for $R = a_0\sqrt{m}$,*

$$\varphi_N(z) \neq 0, \qquad |z| \leq R.$$

More precisely, for some $\gamma_* > a_0^2$,

$$\sup_{|z|\leq R} \left|\varphi_N(z)\varphi_N(-z) - e^{-z^2}\right| \leq e^{-\gamma_* m}.$$

Proof Since $|X_N| \leq N$, φ_N is entire and

$$|\varphi_N(z)| \leq e^{N|\Im z|}.$$

Hence, for

$$\mathcal{F}_N(z) := \varphi_N(z)\varphi_N(-z) - e^{-z^2},$$

the explicit growth bound

$$\log|\mathcal{F}_N(z)| \leq \log 2 + 4NR + 4R^2 \leq \log 2 + 4(Ba_0 + a_0^2)m, \qquad |z| \leq 2R,$$

holds. On the real diameter, Lemma 3.4 gives $\log|\mathcal{F}_N| \leq -\Gamma m$. Apply Lemma 3.1 with radius $2R$ and $\eta = 1/2$ in the upper and lower half-disks. With $\omega_0 := \omega_{1/2}$, this gives

$$\log|\mathcal{F}_N(z)| \leq -\left[\omega_0\Gamma - 4(1-\omega_0)(Ba_0 + a_0^2)\right] m + O(1).$$

Appendix A chooses B and then a_0 so that the bracket is a number $\gamma_* > a_0^2$. Since

$$\inf_{|z|\leq R} |e^{-z^2}| \geq e^{-R^2} = e^{-a_0^2 m},$$

we have the relative estimate

$$\frac{|\mathcal{F}_N(z)|}{|e^{-z^2}|} \leq e^{-(\gamma_* - a_0^2)m}.$$

Since $\gamma_* > a_0^2$, this is less than $1/2$ for all sufficiently large m. Hence

$$|\varphi_N(z)\varphi_N(-z)| \geq \frac{1}{2}|e^{-z^2}| > 0,$$

and therefore $\varphi_N(z) \neq 0$ on the disk. $\square$

3.6 From zero-freeness to Gaussian approximation

Lemma 3.6 (From zero-freeness to Gaussian approximation) *Under the conclusions of Lemma 3.5,*

$$d_{\mathrm{K}}(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \leq \frac{C}{R},$$

where $\mu_N = \mathbb{E}X_N$ and $\sigma_N^2 = \mathrm{Var}(X_N)$. Consequently

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \leq \frac{C}{\sqrt{m}}.$$

Proof The disk $D_R := \{z : |z| < R\}$ is simply connected, and Lemma 3.5 gives $\varphi_N \neq 0$ there. Hence there is a unique analytic logarithm on D_R satisfying $\psi_N(0) = 0$; write

$$\psi_N(z) = \log \varphi_N(z), \qquad \psi_N(0) = 0.$$

The bounded support gives $\Re\psi_N(z) = \log|\varphi_N(z)| \leq NR$. Set

$$H(z) := NR - \psi_N(z), \qquad \Re H(z) \geq 0.$$

After scaling $z = Rw$, the Carathéodory coefficient theorem for functions with nonnegative real part (see, for example, Duren [14]) applied to $H(Rw)/(NR)$ gives, for $\psi_N(z) = \sum_{k\geq 1} a_k z^k$,

$$|a_k| \leq 2NR^{1-k}.$$

Since $a_1 = i\mu_N$ and $a_2 = -\sigma_N^2/2$, for every fixed $0 < \theta < 1$,

$$\left|\psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2}\right| \leq \sum_{k\geq 3} 2NR^{1-k}|t|^k \leq \frac{2N|t|^3}{R^2(1-|t|/R)} \leq C\frac{|t|^3}{R}, \qquad |t| \leq \theta R,$$

because $N/R = B/a_0$ is fixed. Define the cubic remainder

$$E_N(t) := \psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2}.$$

By Lemma 3.3, for all sufficiently large m,

$$|\mu_N| = o(R^{-1}), \qquad \frac{1}{2} \leq \sigma_N^2 \leq \frac{3}{2}, \qquad |E_N(t)| \leq C\frac{|t|^3}{R}.$$

Choose θ so small that $C\theta \leq 1/8$. Then, on $|t| \leq \theta R$,

$$\Re\psi_N(t) = -\frac{\sigma_N^2 t^2}{2} + \Re E_N(t) \leq -\frac{t^2}{4} + \frac{t^2}{8} = -\frac{t^2}{8}.$$

Writing

$$\varphi_N(t) = e^{i\mu_N t - \sigma_N^2 t^2/2} e^{E_N(t)},$$

and using $|e^z - 1| \leq |z|e^{|z|}$ gives Gaussian damping in the difference:

$$\left|\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}\right| \leq C\frac{|t|^3}{R} e^{-ct^2}.$$

Let g_N be the density of $\mathcal{N}(\mu_N, \sigma_N^2)$. The Esseen smoothing inequality in the present normalization (Esseen [15]) is

$$\begin{aligned} d_{\mathrm{K}}(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) &\leq \frac{1}{\pi}\int_{-T}^{T} \left|\frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t}\right| dt + \frac{24\|g_N\|_\infty}{\pi T} \\ &= \frac{1}{\pi}\int_{-T}^{T} \left|\frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t}\right| dt + O(T^{-1}), \end{aligned}$$

where $\|g_N\|_\infty = (\sqrt{2\pi}\sigma_N)^{-1}$ and the last form uses $\sigma_N^2 \geq 1/2$. With $T = \theta R$, the integral is bounded by

$$\frac{C}{R}\int_{-T}^{T} t^2 e^{-ct^2}\, dt \leq \frac{C}{R}\int_{\mathbb{R}} t^2 e^{-ct^2}\, dt = O(R^{-1}),$$

and the cutoff term is also $O(R^{-1})$. The tail lemma gives $d_{\mathrm{K}}(X, X_N) \leq d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(X_N))$ and exponentially small mean and variance shifts. Since the Gaussian distribution function has a bounded density, these shifts contribute $o(R^{-1})$. Thus the displayed bound holds, and the triangle inequality gives the consequence for X. □

Proof of Theorem 1.2 Apply Lemmas 3.2, 3.3, 3.4, 3.5, and 3.6. Since

$$m \asymp \frac{L}{\log L}, \qquad L = \log(1/\Delta),$$

we have

$$\frac{1}{\sqrt{m}} \asymp \sqrt{\frac{\log L}{L}},$$

which is the claimed rate. $\square$

4 Optimality and obstruction results

4.1 A Laguerre lower-bound family

We give the lower-bound construction in full because it identifies the scale in Theorem 1.2. The only analytic input needed for the probability construction is the following uniform Laguerre envelope; its normalization and four-region proof are recorded in Appendix B.

Lemma 4.1 (Uniform Laguerre envelope)

$$C_L := \sup_{m\geq 1,\, x\in\mathbb{R}} \left| x e^{-x^2} L_m^{(1/2)}(3x^2/2) \right| < \infty. \tag{6}$$

Proof The four-region estimate and the conversion from the normalized Laguerre function are given in Appendix B. $\square$

Fix $0 < \epsilon < 1/4$, let ϕ be the standard Gaussian density, and define

$$p_m(x) := x e^{-x^2} L_m^{(1/2)}(3x^2/2), \qquad h_m(x) := \frac{p_m(x)}{C_L}, \qquad f_m(x) := \phi(x)(1+\epsilon h_m(x)).$$

By (6), f_m is a nonnegative probability density. It is normalized because h_m is odd. The change of variables $r = 3x^2/2$ and Laguerre orthogonality show that

$$\int_{\mathbb{R}} x^{2k+1} h_m(x)\phi(x)\, dx = 0, \qquad 0 \leq k < m.$$

In particular, the corresponding variable U_m is centered. Its variance is one because $x^2 h_m(x)\phi(x)$ is odd. To record the generating-function calculation behind the exact MGF, set

$$I_m(s) := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x e^{sx-3x^2/2} L_m^{(1/2)}(3x^2/2)\, dx.$$

The Laguerre generating function gives, for $|t| < 1$,

$$\sum_{m\geq 0} I_m(s) t^m = \frac{(1-t)^{-3/2}}{\sqrt{2\pi}} \int_{\mathbb{R}} x \exp\left(sx - \frac{3x^2}{2(1-t)} \right) dx$$

$$= \frac{s}{3\sqrt{3}} \exp\left(\frac{s^2(1-t)}{6}\right)$$
$$= \frac{s}{3\sqrt{3}} e^{s^2/6} \sum_{m\geq 0} \frac{(-s^2/6)^m}{m!} t^m.$$

Thus $I_m(s) = \frac{s}{3\sqrt{3}} e^{s^2/6}(-s^2/6)^m/m!$. Dividing the perturbation integral by $e^{s^2/2}$ yields the exact MGF

$$M_{U_m}(s) = e^{s^2/2}(1 + u_m(s)), \qquad u_m(s) = \frac{\epsilon}{C_L} \frac{s}{3\sqrt{3}} e^{-s^2/3} \frac{(-s^2/6)^m}{m!}. \tag{7}$$

Therefore, with

$$a_m := \frac{\epsilon}{3\sqrt{3} C_L 6^m m!},$$

we have $|u_m(s)| \leq a_m |s|^{2m+1}$ on the real axis.

Let $J_m = N_m^+ - N_m^-$, where N_m^+ and N_m^- are independent Poisson variables with mean $\lambda_m/2$, independent also of U_m, and set

$$\lambda_m := 96 a_m^2 \rho^{4m-2}, \qquad W_m := \frac{U_m + J_m}{\sqrt{1+\lambda_m}}.$$

The independence gives

$$M_{U_m+J_m}(q) = M_{U_m}(q) M_{J_m}(q), \qquad M_{J_m}(q) = \exp\bigl(\lambda_m(\cosh q - 1)\bigr).$$

Then W_m is centered and variance one. If $q = s/\sqrt{1+\lambda_m}$, its reflected defect is exactly

$$R_{W_m}(s) = \log(1 - u_m(q)^2) + 2\lambda_m \left(\cosh q - 1 - \frac{q^2}{2}\right). \tag{8}$$

For all sufficiently large m, $|u_m(q)| \leq 1/2$ on $|s| \leq \rho$. Using $-\log(1-v) \leq 2v$ for $0 \leq v \leq 1/4$ and $\cosh q - 1 - q^2/2 \geq q^4/24$, the choice of λ_m gives

$$R_{W_m}(s) \geq \frac{\lambda_m q^4}{16} \geq 0.$$

The logarithmic term in (8) is nonpositive, so the upper Taylor bound gives

$$R_{W_m}(s) \leq \frac{\lambda_m \cosh(\rho) q^4}{12}.$$

If also $\lambda_m \leq 1$, evaluation at $s = \rho$ yields

$$\frac{\lambda_m \rho^4}{64} \leq \Delta_m := \sup_{|s|\leq\rho} R_{W_m}(s) \leq \frac{\lambda_m \rho^4 \cosh(\rho)}{12}.$$

Thus $\Delta_m \asymp \lambda_m$ and Stirling's formula gives

$$\log \frac{1}{\Delta_m} = 2m \log m + O(m).$$

Consequently,

$$\log \log(1/\Delta_m) = \log m + \log \log m + O(1), \qquad \frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)} \asymp \frac{1}{m},$$

and hence

$$m^{-1/2} \asymp \sqrt{\frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)}}.$$

The fixed exponential envelope is uniform: $f_m \le (1+\epsilon)\phi$, while

$$\mathbb{E} e^{b|J_m|} \le 2e^{\lambda_m(\cosh b - 1)}$$

and $\sup_m \lambda_m < \infty$. Finally, the half-axis discrepancy is explicit. With $r = 3x^2/2$, one has $x\,dx = dr/3$ on $(0,\infty)$, and hence

$$\int_0^\infty h_m(x)\phi(x)\,dx = \frac{1}{3C_L\sqrt{2\pi}} \int_0^\infty e^{-r} L_m^{(1/2)}(r)\,dr.$$

The Laguerre generating function gives

$$\sum_{m\ge 0} t^m \int_0^\infty e^{-r} L_m^{(1/2)}(r)\,dr = (1-t)^{-3/2} \int_0^\infty e^{-r/(1-t)}\,dr = (1-t)^{-1/2}.$$

Comparing coefficients yields $\int_0^\infty e^{-r} L_m^{(1/2)}(r)\,dr = (1/2)_m/m!$. Therefore

$$\int_0^\infty h_m(x)\phi(x)\,dx = \frac{1}{3C_L\sqrt{2\pi}} \frac{(1/2)_m}{m!},$$

and $(1/2)_m/m! \asymp m^{-1/2}$ gives $|F_{U_m}(0) - 1/2| \asymp m^{-1/2}$. Moreover,

$$|F_{U_m+J_m}(0) - F_{U_m}(0)| \le \mathbb{P}(J_m \ne 0) \le \lambda_m = o(m^{-1/2}).$$

The normalization in W_m leaves the threshold zero unchanged. Consequently

$$d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) \ge c m^{-1/2}.$$

Together with $\log(1/\Delta_m) = 2m\log m + O(m)$, this proves Theorem 1.3 and rules out every uniform power-law modulus $C\Delta^\alpha$, $\alpha > 0$.

4.2 Failure of total-variation stability

Throughout this subsection, $d_{\mathrm{TV}}(\mu,\nu) := \sup_A |\mu(A) - \nu(A)|$.

Proposition 4.2 (Strong-distance obstruction) *There is no uniform total-variation stability theorem under the assumptions of Theorem 1.2: a standardized Poisson lattice family can satisfy a fixed exponential-integrability bound and have reflected defect tending to zero while its total-variation distance from every absolutely continuous Gaussian law equals one.*

Let $N_\lambda \sim \mathrm{Poisson}(\lambda)$ and $X_\lambda = (N_\lambda - \lambda)/\sqrt{\lambda}$. Its centered cumulant generating function is

$$K_\lambda(s) = \lambda\left(e^{s/\sqrt{\lambda}} - 1 - \frac{s}{\sqrt{\lambda}}\right).$$

Consequently, uniformly on every fixed compact s-interval,

$$\begin{aligned} R_{X_\lambda}(s) &= K_\lambda(s) + K_\lambda(-s) - s^2 \\ &= 2\lambda\left(\cosh(s/\sqrt{\lambda}) - 1 - \frac{s^2}{2\lambda}\right) = \frac{s^4}{12\lambda} + O_\rho(\lambda^{-2}). \end{aligned}$$

For fixed $b > 0$, the inequality $e^u - 1 - u \le u^2 e^{|u|}/2$ gives

$$\mathbb{E}e^{b|X_\lambda|} \le \mathbb{E}e^{bX_\lambda} + \mathbb{E}e^{-bX_\lambda} \le 2\exp\left(\frac{b^2}{2}e^{b/\sqrt{\lambda}}\right),$$

so the family has a fixed exponential-integrability bound for each fixed b. Nevertheless X_λ is lattice-valued, whereas $\mathcal{N}(0,1)$ is absolutely continuous, so

$$d_{\mathrm{TV}}(\mathcal{L}(X_\lambda), \mathcal{N}(0,1)) = 1.$$

This is a qualitative boundary phenomenon and is independent of the Kolmogorov lower bound above.

5 Sample variance

Let $X_1, \ldots, X_n$ be i.i.d. copies of a centered variance-one random variable X, let $\bar{X} = n^{-1}\sum_i X_i$, and set

$$Q_n = \sum_{i=1}^n (X_i - \bar{X})^2.$$

For $r \ge 1$, let V be uniform on S^{r-1}. Recall the spherical transform

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1}.$$

Define

$$\mathfrak{D}_{r,\tau}(Q) := \sup_{|t|\le\tau} \left| e^{-t^2/2}\mathbb{E}\Psi_r(t\sqrt{Q}) - 1\right|.$$

This is the spherical Laplace observable associated with a squared radial projection; it is not a probability metric on the law of Q.

The exact sample-variance consequence of the quadratic-form theorem is the following.

Corollary 5.1 (Exact sample-variance characterization under dominance) *Let $n \geq 2$ and let $X_1, \dots, X_n$ be i.i.d., centered, and variance one. If M_X is finite near the origin,*

$$M_X(s)M_X(-s) \geq e^{s^2}$$

there, and

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 \sim \chi^2_{n-1},$$

then $X_i \sim \mathcal{N}(0,1)$.

The quantitative statement is the next theorem.

Theorem 5.2 (Sample-variance stability under spherical Laplace discrepancy) *Fix $n \geq 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There are constants $C < \infty$ and $\varepsilon_0 > 0$, depending only on (n, τ, K, ρ'), such that the following holds. Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one X with $\mathbb{E}e^{2\tau|X|} \leq K$, and assume $R_X(s) \geq 0$ for $|s| \leq \tau$. If*

$$Q_n = \sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \mathfrak{D}_{n-1,\tau}(Q_n) \leq \varepsilon \leq \varepsilon_0,$$

then

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \leq C\sqrt{\frac{\log\log(1/\varepsilon)}{\log(1/\varepsilon)}}.$$

The discrepancy-to-defect transfer is as follows. Let $r = n-1$, let $P = I_n - n^{-1}\mathbf{1}\mathbf{1}^\mathsf{T}$, and choose $B \in \mathbb{R}^{r\times n}$ with $B^\mathsf{T}B = P$ and $BB^\mathsf{T} = I_r$. If V is uniform on S^{r-1}, then every column b_i of B has norm

$$q := \|b_i\| = \sqrt{\frac{n-1}{n}},$$

and $\langle b_i, V\rangle$ has the same distribution as qV_1. Put $a = \tau q$ and define

$$D_X(s) := \frac{1}{2}R_X(s) = \frac{1}{2}\log\frac{M_X(s)M_X(-s)}{e^{s^2}}.$$

Under dominance, $D_X \geq 0$ on $[-\tau, \tau]$.

The exact spherical transform gives, for $|t| \leq \tau$,

$$e^{-t^2/2}\mathbb{E}\Psi_r(t\sqrt{Q_n}) = \mathbb{E}_V \exp\left(\sum_{i=1}^{n}\left[\log M_X(t\langle b_i, V\rangle) - \frac{t^2\langle b_i, V\rangle^2}{2}\right]\right). \tag{9}$$

Jensen's inequality, exchangeability, and the symmetry of V_1 therefore imply

$$n\mathbb{E}D_X(aV_1) \leq \log\bigl(1 + \mathfrak{D}_{n-1,\tau}(Q_n)\bigr). \tag{10}$$

Equation (10) connects the spherical observable to one-dimensional reflection control.

For the pointwise transfer, the density of V_1 is

$$f_n(v) = \frac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma((n-2)/2)}(1-v^2)^{(n-4)/2}, \qquad -1 < v < 1.$$

Fix $0 < \rho' < a$ and choose $0 < h < a - \rho'$. On $[-(\rho'+h), \rho'+h]$ the scaled density of aV_1 is bounded below by

$$m_h := \frac{1}{a}\frac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma((n-2)/2)}\left(1 - \left(\frac{\rho'+h}{a}\right)^2\right)^{\max(n-4,0)/2}.$$

The exponential envelope implies that D_X is Lipschitz there. One may take

$$L_h := \frac{K}{e\tau} + \rho' + h,$$

because $M_X(s) \geq 1$ and $|x|e^{\tau|x|} \leq (e\tau)^{-1}e^{2\tau|x|}$. If $M_* := \sup_{|s|\leq\rho'} D_X(s)$, choose $s_0 \in [-\rho', \rho']$ with $D_X(s_0) = M_*$. The function D_X is even, nonnegative, and satisfies $D_X(0) = 0$. Since it is L_h-Lipschitz on the enlarged interval,

$$D_X(s) \geq \bigl(M_* - L_h|s - s_0|\bigr)_+.$$

Moreover, $|s_0| \geq M_*/L_h$, so the analogous tent centered at $-s_0$ has disjoint interior from the other one. Integrating the two tents against the density lower bound gives

$$2m_h I_{L_h,h}(M_*) \leq \frac{1}{n}\log\bigl(1 + \mathfrak{D}_{n-1,\tau}(Q_n)\bigr),$$

where

$$I_{L,h}(M) := \begin{cases} M^2/L, & M \leq Lh, \\ 2hM - Lh^2, & M > Lh. \end{cases}$$

When $M \leq Lh$, the integral is the area of a triangle of base $2M/L$; when $M > Lh$, it is the trapezoid obtained after the tent reaches the edge of the available interval. Writing $G(y; L, h) = \sqrt{Ly}$ when $y \leq Lh^2$ and $G(y; L, h) = y/(2h) + Lh/2$ otherwise, we obtain

$$\sup_{|s|\leq\rho'} D_X(s) \leq \inf_{0<h<a-\rho'} G\left(\frac{\log(1 + \mathfrak{D}_{n-1,\tau}(Q_n))}{2nm_h}; L_h, h\right). \tag{11}$$

In particular, for fixed n, τ, K, ρ', the right-hand side is $O(\sqrt{\mathfrak{D}_{n-1,\tau}(Q_n)})$ as the discrepancy tends to zero. The strict interior restriction is essential when $n \geq 5$, because the spherical density vanishes at the endpoints.

Applying (11) to the sample-variance hypothesis gives

$$\Delta_{\rho'}(X) \le C_1\sqrt{\varepsilon},$$

for all sufficiently small ε, where $C_1 = C_1(n, \tau, K, \rho')$; here we used $R_X = 2D_X$. Set $\Delta = C_1\sqrt{\varepsilon}$. Then

$$\log(1/\Delta) = \tfrac{1}{2}\log(1/\varepsilon) - \log C_1 \asymp \log(1/\varepsilon), \qquad \log\log(1/\Delta) = \log\log(1/\varepsilon) + O(1).$$

Theorem 1.2, with radius ρ' and the fixed exponential envelope, therefore yields the displayed rate in Theorem 5.2. In the exact case, Corollary 5.1 gives Gaussianity under the same dominance condition. For $n = 2$, the exact corollary is covered by the rank-one case of Theorem 1.1. Indeed, $A = I_2 - \frac{1}{2}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ has rank one and both diagonal entries are positive. Thus the discrepancy-to-defect argument is a quantitative extension of the exact spherical-Jensen mechanism.

Two questions seem particularly natural. First, can the exponential-moment assumption in the reflection theorem be weakened without changing the optimal modulus? Second, under what additional uniform-integrability hypothesis can the spherical Laplace discrepancy in the sample-variance application be replaced by a standard probability metric? It would also be interesting to develop a corresponding quantitative theory for general positive-semidefinite quadratic forms with non-identically distributed coordinates.

A Parameter selection for the sharp upper bound

This appendix records a non-circular order of choices for the five-step proof. Fix (ρ, τ, K) and an interior analytic radius. Let $\omega_0 > 0$ be the lower bound for the harmonic measure of the real diameter in the smaller half-disk used in the propagation step. Choose $A > 8$ and then choose $c_0 > 0$ so small that

$$(A+3)c_0 < \omega_0.$$

With $m = \lfloor c_0 L / \log L \rfloor$, this makes the fixed-disk Cauchy estimate dominate all factorial losses in the moment-transfer lemma.

Fix a finite target exponent $\Gamma > 4$. Choose the truncation factor B sufficiently large that the tail-lifting estimate has exponent $\beta_B > 4\Gamma$. After B is fixed, let C_T denote the constant in the real-axis Taylor remainder and choose $a_0 > 0$ so small that

$$C_T a_0^2 < e^{-2\Gamma}, \qquad 4(1-\omega_0)(Ba_0 + a_0^2) < \frac{\omega_0 \Gamma}{2}, \qquad a_0^2 < \frac{\omega_0 \Gamma}{4}, \qquad -\log(C_T a_0^2) > 4\Gamma.$$

The two Taylor-remainder contributions are then bounded by $e^{-\Gamma m}$ after absorbing fixed multiplicative constants, while the truncation contribution is also bounded by $e^{-\Gamma m}$ because $\beta_B > 4\Gamma$. Thus the real-axis estimate in Lemma 3.4 uses the target exponent Γ, while the complex propagation estimate gives

$$\gamma_* := \omega_0 \Gamma - 4(1-\omega_0)(Ba_0 + a_0^2) \geq \frac{\omega_0 \Gamma}{2} > a_0^2.$$

Finally choose the fixed Esseen fraction $0 < \theta < 1/4$ so that the cubic remainder in the logarithmic expansion is at most one quarter of the Gaussian quadratic damping on $|t| \leq \theta R$. These choices are made in the order $A \to c_0 \to \Gamma \to B \to a_0 \to \theta$, and no later choice affects an earlier inequality.

B A uniform Laguerre estimate

For $\alpha = 1/2$, use the normalized Laguerre function

$$\mathcal{L}_m^{(1/2)}(r) = a_m L_m^{(1/2)}(r) e^{-r/2} r^{1/4}, \qquad a_m = \sqrt{\frac{m!}{\Gamma(m+3/2)}}, \qquad \nu = 4m+3.$$

Proposition 3.2 of Imekraz, Robert, and Thomann [16] gives constants $C, c > 0$, independent of m and r, such that

$$|\mathcal{L}_m^{(1/2)}(r)| \leq C(r\nu)^{1/4} \quad (0 \leq r \leq \nu^{-1}),$$

$$|\mathcal{L}_m^{(1/2)}(r)| \leq C(r\nu)^{-1/4} \quad (\nu^{-1} \leq r \leq \nu/2),$$

$$|\mathcal{L}_m^{(1/2)}(r)| \leq C\nu^{-1/4}\big(\nu^{1/3} + |\nu - r|\big)^{-1/4} \quad (\nu/2 \leq r \leq 3\nu/2),$$

and

$$|\mathcal{L}_m^{(1/2)}(r)| \le Ce^{-cr} \quad (r \ge 3\nu/2).$$

Since

$$Q_m(r) := \sqrt{r}\, e^{-r/2}|L_m^{(1/2)}(r)| = a_m^{-1} r^{1/4}|\mathcal{L}_m^{(1/2)}(r)|, \qquad a_m^{-1} \asymp m^{1/4} \asymp \nu^{1/4},$$

the four regions can be checked without hiding the powers of m.

In Region I, $0 \le r \le \nu^{-1}$, the estimate gives

$$Q_m(r) \le C\nu^{1/4} r^{1/4}(r\nu)^{1/4} = C(r\nu)^{1/2} \le C.$$

In Region II, $\nu^{-1} \le r \le \nu/2$, it gives

$$Q_m(r) \le C\nu^{1/4} r^{1/4}(r\nu)^{-1/4} = C.$$

In the turning region, $\nu/2 \le r \le 3\nu/2$, we obtain

$$Q_m(r) \le Cr^{1/4}\big(\nu^{1/3} + |\nu - r|\big)^{-1/4} \le C\nu^{1/6}.$$

Multiplication by the remaining factor in the desired envelope gives

$$e^{-r/6} Q_m(r) \le C\nu^{1/6} e^{-\nu/12} \le C.$$

Finally, in the outer region $r \ge 3\nu/2$, the exponential estimate yields

$$Q_m(r) \le C\nu^{1/4} r^{1/4} e^{-cr} \le C \text{ poly}(r, \nu) e^{-cr}.$$

After multiplication by $e^{-r/6}$ this is uniformly bounded. Hence

$$\boxed{\sup_{m\ge 1,\, r\ge 0} \sqrt{r}\, e^{-2r/3}|L_m^{(1/2)}(r)| < \infty.}$$

The case $m = 0$ is immediate from $L_0^{(1/2)}(r) = 1$. Substituting $r = 3x^2/2$ converts this bound into the constant C_L used in Section 4.

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